

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_weighted_replay.
- Best TSPLIB holdout gap: phase6_diversity_weighted_replay.
- Best synthetic holdout gap: phase6_failure_replay.

Run Metadata

- run_name: run_20260519_020811_l
- started_at_local: 2026-05-19 02:08:11
- finished_at_local: 2026-05-19 02:31:36
- duration_hhmm: 00:23
- duration_seconds: 1404.804
- seed_offset: 11000
- replicate_label: l
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.0	0.772439	0.56	-0.032242
phase6_random_replay	random	score_only	False	0.114822	0.003941	0.074502	0.236772	0.193042	0.19457	0.804341	0.76	0.032453
phase6_failure_replay	failure	score_only	False	0.155899	0.0	0.099208	0.171206	0.269686	0.212826	0.0	0.76	0.0

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.148009	0.0	0.094188	0.236772	0.188451	0.19457	0.745793	0.76	0.013215
phase6_stratified_random_replay	stratified_random	score_only	False	0.146551	0.0	0.09326	0.236772	0.146727	0.221162	0.841642	0.76	0.010515
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.100221	0.0	0.063777	0.236772	0.170475	0.130606	0.787962	0.76	-0.001085
phase6_residual_failure_replay	residual_failure	score_only	False	0.191306	0.001628	0.122332	0.121572	0.268644	0.449508	0.775017	0.76	-0.002285
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.186646	0.010343	0.122536	0.171206	0.267501	0.171578	0.0	0.76	0.0
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.0	0.56	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.772439, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency -0.032242 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.404343, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.003025}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.133545}],

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{"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.322373, "family": "lin", "name": "lin105",  
"optimality_gap": 0.328813, "residual_gap": 0.00644}]
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phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.114822, synthetic 0.003941, combined 0.074502.
- Accepted-epoch count 3, mean accepted code novelty 0.804341, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.193042, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133281, mean selected residual gap 0.023178.
- Adaptation efficiency 0.032453 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.114822 across 7 instances; family means: ch=0.141003, kroD=0.104067, pcb=0.185454, pr=0.036113, rd=0.102781, st=0.093333.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3474, "expected_gap": 0.40442, "family": "a", "name": "a280", "optimality_gap": 0.347034, "residual_gap": -0.057386}, {"best_known_cost": 7542, "cost": 9644, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.278706, "residual_gap": 0.045638}, {"best_known_cost": 14379, "cost": 17603, "expected_gap": 0.321803, "family": "lin", "name": "lin105", "optimality_gap": 0.224216, "residual_gap": -0.097587}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.155899, synthetic 0.0, combined 0.099208.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.269686, size bias 0.120322, and failure concentration 0.212826.
- Replay selection diversity 0.196656, mean selected expected gap 0.322557, mean selected residual gap -0.013127.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.155899 across 7 instances; family means: ch=0.131091, kroD=0.144736, pcb=0.2172, pr=0.18495, rd=0.18445, st=0.097778.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3011, "expected_gap": 0.406824, "family": "a", "name": "a280", "optimality_gap": 0.167507, "residual_gap": -0.239317}, {"best_known_cost": 7542, "cost": 8593, "expected_gap": 0.235826, "family": "berlin", "name": "berlin52", "optimality_gap": 0.139353, "residual_gap": -0.096473}, {"best_known_cost": 14379, "cost": 16167, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.124348, "residual_gap": -0.197163}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.148009, synthetic 0.0, combined 0.094188.

- Accepted-epoch count 3, mean accepted code novelty 0.745793, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.188451, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132484, mean selected residual gap 0.003609.
- Adaptation efficiency 0.013215 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.148009 across 7 instances; family means: ch=0.112974, kroD=0.224617, pcb=0.199772, pr=0.195934, rd=0.154235, st=0.035556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3094, "expected_gap": 0.404498, "family": "a", "name": "a280", "optimality_gap": 0.19969, "residual_gap": -0.204808}, {"best_known_cost": 14379, "cost": 16228, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.12859, "residual_gap": -0.192921}, {"best_known_cost": 7542, "cost": 8338, "expected_gap": 0.243888, "family": "berlin", "name": "berlin52", "optimality_gap": 0.105542, "residual_gap": -0.138346}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.146551, synthetic 0.0, combined 0.09326.
- Accepted-epoch count 3, mean accepted code novelty 0.841642, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.146727, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236082, mean selected residual gap -0.035819.
- Adaptation efficiency 0.010515 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.146551 across 7 instances; family means: ch=0.133548, kroD=0.171832, pcb=0.231006, pr=0.224817, rd=0.110367, st=0.020741.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3572, "expected_gap": 0.405661, "family": "a", "name": "a280", "optimality_gap": 0.385033, "residual_gap": -0.020628}, {"best_known_cost": 629, "cost": 805, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.279809, "residual_gap": 0.023529}, {"best_known_cost": 14379, "cost": 17409, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.210724, "residual_gap": -0.11055}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.100221, synthetic 0.0, combined 0.063777.
- Accepted-epoch count 2, mean accepted code novelty 0.787962, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.170475, size bias 8.3e-05, and failure concentration 0.130606.
- Replay selection diversity 0.263228, mean selected expected gap 0.180927, mean selected residual gap 0.000344.

- Adaptation efficiency -0.001085 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.100221 across 7 instances; family means: ch=0.114427, kroD=0.111064, pcb=0.214424, pr=0.036113, rd=0.051833, st=0.059259.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3273, "expected_gap": 0.404188, "family": "a", "name": "a280", "optimality_gap": 0.269097, "residual_gap": -0.135091}, {"best_known_cost": 629, "cost": 684, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.08744, "residual_gap": -0.166614}, {"best_known_cost": 14379, "cost": 15352, "expected_gap": 0.317296, "family": "lin", "name": "lin105", "optimality_gap": 0.067668, "residual_gap": -0.249628}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.191306, synthetic 0.001628, combined 0.122332.
- Accepted-epoch count 2, mean accepted code novelty 0.775017, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.121572, mean hardness 0.268644, size bias 0.055669, and failure concentration 0.449508.
- Replay selection diversity 0.129326, mean selected expected gap 0.252916, mean selected residual gap 0.049495.
- Adaptation efficiency -0.002285 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.191306 across 7 instances; family means: ch=0.174453, kroD=0.309336, pcb=0.253003, pr=0.228497, rd=0.106068, st=0.093333.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3657, "expected_gap": 0.405196, "family": "a", "name": "a280", "optimality_gap": 0.417991, "residual_gap": 0.012795}, {"best_known_cost": 629, "cost": 830, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.319555, "residual_gap": 0.057552}, {"best_known_cost": 14379, "cost": 18335, "expected_gap": 0.328729, "family": "lin", "name": "lin105", "optimality_gap": 0.275123, "residual_gap": -0.053606}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.186646, synthetic 0.010343, combined 0.122536.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.267501, size bias 0.120322, and failure concentration 0.171578.
- Replay selection diversity 0.201543, mean selected expected gap 0.307782, mean selected residual gap -0.036014.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.186646 across 7 instances; family means: ch=0.099378, kroD=0.24251, pcb=0.273662, pr=0.340869, rd=0.225537, st=0.025185.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3463, "expected_gap": 0.403955, "family": "a", "name": "a280", "optimality_gap": 0.342769, "residual_gap": -0.061186}, {"best_known_cost": 7542, "cost": 9442, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.251923, "residual_gap": 0.006816}, {"best_known_cost": 14379, "cost": 17366, "expected_gap": 0.328131, "family": "lin", "name": "lin105", "optimality_gap": 0.207734, "residual_gap": -0.120397}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.303068, mean selected residual gap 0.04692.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.407212, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.005894}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.127186}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.329634, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": -0.000821}]

Judge Appendix

TSP Benchmark Suite Replay-Aware Condition Analysis

Summary of Best Conditions

- ****Best Holdout (TSPLIB) Condition:**** phase6_diversity_weighted_replay
- Final TSPLIB gap: 0.100221 (lowest across conditions)
- ****Best Synthetic Holdout Condition:**** phase6_failure_replay
- Synthetic gap: 0.0 (best possible)
- ****Best Transfer Condition:**** phase6_diversity_weighted_replay
- Transfer gap: 0.063777 (lowest transfer gap)

Comparative Performance Overview

Condition Name	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Replay Mode	Selection Mode
phase6_no_replay	0.213387	0.064916	0.159398	none	score_only
phase6_random_replay	0.114822	0.003941	0.074502	random	score_only
phase6_failure_replay	0.155899	0.0	0.099208	failure	score_only
phase6_random_replay_compression	0.148009	0.0	0.094188	random (compression)	novelty_gate
phase6_stratified_random_replay	0.146551	0.0	0.093260	stratified_random	score_only
phase6_diversity_weighted_replay	**0.100221**	0.0	**0.063777**	diversity_weighted	score_only
phase6_residual_failure_replay	0.191306	0.001628	0.122332	residual_failure	score_only
phase6_diversity_failure_replay	0.186646	0.010343	0.122536	diversity_failure	score_only
phase6_diversity_failure_replay_compression	0.213387	0.064916	0.159398	diversity_failure (compression)	novelty_gate

- ****Note:**** Lower gaps represent better performance (closer to optimal).

In-Depth Analysis

1. Optimality Gaps & Transfer Performance

- ****Held-out TSPLIB Gap (main transfer metric):****
- Best achieved by phase6_diversity_weighted_replay (0.100221), which is a substantial improvement over no replay (0.213387).
- Other replay strategies yield moderate improvements but do not match the diversity-weighted replay's performance.
- ****Synthetic Holdout Gap:****
- Failure replay and variants (including stratified random) achieve zero or near-zero synthetic gap, signaling strong adaptation.
- Raw random replay also has very low synthetic gap (0.003941), indicating good fitting on synthetic challenges.
- ****Transfer Gap:****
- Lowest transfer gap observed for phase6_diversity_weighted_replay at 0.063777.
- Others, like failure replay and random replay, have moderate transfer gaps (~0.07-0.1).
- No replay and diversity failure replay conditions have larger transfer gaps nearing 0.15-0.16.

2. Replay Archive Diversity & Hardness

Condition	Archive Diversity	Archive Hardness	Archive Size Bias
phase6_no_replay	0.240179	0.212361	0.0
phase6_random_replay	0.240179	0.205529	0.0
phase6_failure_replay	0.174718	0.333292	0.124979
phase6_random_replay_compression	0.240179	0.194479	0.0
phase6_stratified_random_replay	0.240179	0.159287	0.0
phase6_diversity_weighted_replay	0.240179	0.194959	0.0
phase6_residual_failure_replay	0.19662	0.334688	0.12889
phase6_diversity_failure_replay	0.174718	0.303871	0.124979
phase6_diversity_failure_replay_compression	0.174718	0.34902	0.124979

- Diversity-weighted replay maintains **high archive diversity (0.24)** with **moderate hardness (~0.20)** and no size bias.
- Failure-based replays tend to have **lower diversity (~0.17-0.20)**, **higher hardness (~0.3-0.35)**, and notable positive size bias (~0.12).
- Random replay preserves diversity better than failure replays but with lower hardness.
- Compression pressure reduces archive size in some conditions but does not strongly affect diversity in diversity-weighted replay.

3. Replay Failure Concentration & Selection Diversity

Condition	Mean Replay Failure Concentration	Mean Replay Selection Diversity	Mean Selected Expected Gap	Mean Selected Residual Gap
phase6_no_replay	0.0	0.0	0.0	0.0
phase6_random_replay	0.19457	0.262174	0.133281	0.023178
phase6_failure_replay	0.212826	0.196656	0.322557	-0.013127
phase6_random_replay_compression	0.19457	0.262174	0.132484	0.003609
phase6_stratified_random_replay	0.221162	0.187367	0.236082	-0.035819
phase6_diversity_weighted_replay	0.130606	0.263228	0.180927	0.000344
phase6_residual_failure_replay	0.449508	0.129326	0.252916	0.000344
phase6_diversity_failure_replay	0.171578	0.201543	0.307782	-0.036014
phase6_diversity_failure_replay_compression	0.17024	0.199284	0.303068	0.04692

- Diversity-weighted replay achieves **lower failure concentration (~0.13)** and **higher selection diversity (~0.26)**.
- Residual failure replay exhibits **high failure concentration (~0.45)** and low selection diversity (~0.13), likely leading to overfitting narrow failure modes.

- Failure replay has moderate to high failure concentration (~0.21), intermediate diversity.
- Random and stratified replay show moderate failure concentration (~0.19-0.22) and higher diversity (~0.18-0.26).

4. Code Novelty vs Transfer

- phase6_diversity_weighted_replay has **mean code novelty ~0.79** with best transfer.
- phase6_failure_replay has **zero code novelty** but excellent synthetic holdout and moderate transfer gains.
- Some diversity or compression modes exhibit **higher code novelty (~0.8-0.84)** with corresponding transfer benefits.
- Conditions with zero code novelty generally have lower transfer performance.

Indicates **transfer improvements can be achieved both with and without high code novelty** depending on replay strategy.

5. Replay Modes Summary

Replay Type	Archive Behavior & Transfer Highlights
No Replay	High TSPLIB gap, moderate synthetic gap, no replay dynamics
Random Replay	Maintains high archive diversity, lower hardness, improved transfer vs no replay
Failure Replay	Lower diversity, higher hardness and size bias, 0 synthetic gap, moderate transfer
Random Replay with Compression	Maintains archive diversity with compression, similar transfer as failure replay
Stratified Random Replay	Slightly lower archive hardness, maintains diversity, good transfer
Diversity-Weighted Replay	Highest diversity, moderate hardness, best TSPLIB and transfer gaps
Residual Failure Replay	High hardness and failure concentration, higher TSPLIB gaps, less efficient adaptation
Diversity Failure Replay	Moderate diversity, higher failure concentration, intermediate transfer
Diversity Failure Replay with Compression	Similar to Diversity Failure Replay but with compression

Conservative Interpretation & Recommendations

- **Best transfer and holdout TSPLIB performance come from phase6_diversity_weighted_replay, which balances archive diversity and hardness, while maintaining low failure concentration and high replay selection diversity.**
- **Failure replay (raw or residual) can achieve zero synthetic gap but tends to have higher failure concentration and lower archive diversity, potentially causing overfitting and worse TSPLIB transfer performance.**
- **Compression-aware replay maintains diversity with slightly lower hardness, offering competitive transfer.**
- **Random-based replay types improve transfer moderately over no replay, benefiting from broad coverage and selection diversity.**
- **Notably, failure replay shows zero code novelty but improved synthetic and transfer gaps, indicating transfer gains can be realized without increased code novelty.**

- ****Thus, code novelty metrics alone are insufficient indicators of algorithmic improvement; transfer gaps provide stronger evidence.****
- ****Residual failure replay shows signs of failure concentration and over-specialization, leading to weaker adaptation despite hardness.****
- ****Diversity-weighted replay carefully manages replay instances to focus on hard but diverse cases, enabling robust transfer.****

Summary

Aspect	Conclusion
Transfer Efficacy	Highest under diversity-weighted replay, followed by random and stratified replay. Failure-based replays are mixed.
Replay Diversity	Diversity-weighted and random replay maintain higher archive and selection diversity, linked to better generalization.
Failure Concentration	Lower in diversity-weighted replay; high in residual failure replay correlates with lower transfer.
Code Novelty	Not strongly correlated with transfer gains; failure replay shows no novelty but some transfer improves.
Compression Effects	Compression-aware random replay sustains diversity and transfer; compression impact depends on replay strategy.

Final Note

****Prioritize phase6_diversity_weighted_replay for best transfer and generalization with balanced archive diversity and failure concentration. Failure replay provides excellent synthetic fitting but risks over-concentration and reduced generalization. Code novelty alone does not explain transfer improvements.****